

Caso_3_ML_2026__Gabriela_Fonseca_Joseph_Doqueresana_Nicolas_Bot

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Entregable 3

K-means y PCA

Métodos de ensamble

Aprendizaje Automático de Máquina

Pregrado MACC 2026-1

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1 Indicaciones

- Para entregar en grupo de dos o tres integrantes (únicamente)
- **Importante:** Cargar en la plataforma *e-aulas* los **dos** archivos: *.ipynb* y *.pdf*
- Fecha de entrega el día **miércoles 6 de Mayo 11:59 p.m.**
- Nombrar los trabajos de la siguiente manera:

Caso_3_ML_2026_(nombre 1-nombre 2-nombre 3)

- En el notebook, indicar el nombre de los integrantes del grupo.

Nota: El incumplimiento de alguna de las condiciones anteriores tiene una penalidad de 0.5 en la nota del entregable.

2 1. PCA

El objetivo de este ejercicio es agrupar países basándose en diversas características numéricas, como indicadores económicos, factores de salud y parámetros sociales. Este es un problema de aprendizaje no supervisado porque no se cuenta con categorías o etiquetas predefinidas para los países.

https://raw.githubusercontent.com/Fabian830348/Bases_Datos/refs/heads/master/datos_paises.csv

Las variables significan lo siguiente:

- **País:** Nombre del país.
- **Mortalidad infantil:** Muerte de niños menores de 5 años por cada 1000 - nacidos vivos.

- **Exportaciones:** Exportaciones de bienes y servicios per cápita. Expresadas como porcentaje del PIB per cápita.
- **Salud:** Gasto total en salud per cápita. Expresado como porcentaje del PIB per cápita.
- **Importaciones:** Importaciones de bienes y servicios per cápita. Expresadas como porcentaje del PIB per cápita.
- **Ingresos:** Ingreso neto anual por persona.
- **Inflación:** Medida de la tasa de crecimiento anual del PIB total.
- **Esperanza de vida:** Número promedio de años que viviría un recién nacido si los patrones actuales de mortalidad permanecen iguales.
- **Fertilidad total:** Número de hijos que tendría cada mujer si las tasas actuales de fertilidad por edad permanecen iguales.
- **PIB per cápita (GDPP):** Producto Interno Bruto por persona. Calculado como el PIB total dividido entre la población total.

1. Cargar y explorar los datos
2. Hacer el análisis de PCA. Identificar las componentes.
3. Hacer el análisis de Biplot.
4. Concluya

2.1 1.1 Librerías

```
[1]: import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
import plotly.io as pio # Import plotly.io
pio.renderers.default = 'colab' # Set default renderer for Colab

from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import silhouette_score, silhouette_samples
from sklearn.cluster import KMeans
from sklearn.neighbors import NearestNeighbors
from sklearn.metrics import pairwise_distances_argmin_min

from yellowbrick.cluster import KElbowVisualizer

import warnings
warnings.filterwarnings("ignore")
```

```
[2]: #para importar la librería pandas.
import pandas as pd
import numpy as np
```

```
# librerías para graficar
import matplotlib.pyplot as plt
import plotly.express as px
import seaborn as sns

# estandarizar los datos
from sklearn.preprocessing import StandardScaler
# importamos la librería para pca clásica
from sklearn.decomposition import PCA
# importamos el escalador estándar
from sklearn.preprocessing import StandardScaler
```

```
[3]: import plotly.io as pio
pio.renderers.default = 'colab' # Set default renderer for Colab to ensure
↳ inline display
```

2.2 1.2 Cargar y explorar los datos

```
[4]: url = "https://raw.githubusercontent.com/Fabian830348/Bases_Datos/refs/heads/
↳ master/datos_paises.csv"
df = pd.read_csv(url)
df.head()
```

```
[4]:
```

	country	child_mort	exports	health	imports	income	\
0	Afghanistan	90.2	10.0	7.58	44.9	1610	
1	Albania	16.6	28.0	6.55	48.6	9930	
2	Algeria	27.3	38.4	4.17	31.4	12900	
3	Angola	119.0	62.3	2.85	42.9	5900	
4	Antigua and Barbuda	10.3	45.5	6.03	58.9	19100	

	inflation	life_expec	total_fer	gdpp
0	9.44	56.2	5.82	553
1	4.49	76.3	1.65	4090
2	16.10	76.5	2.89	4460
3	22.40	60.1	6.16	3530
4	1.44	76.8	2.13	12200

```
[5]: df.shape
```

```
[5]: (167, 10)
```

```
[6]: df.describe()
```

```
[6]:
```

	child_mort	exports	health	imports	income	\
count	167.000000	167.000000	167.000000	167.000000	167.000000	
mean	38.270060	41.108976	6.815689	46.890215	17144.688623	

std	40.328931	27.412010	2.746837	24.209589	19278.067698
min	2.600000	0.109000	1.810000	0.065900	609.000000
25%	8.250000	23.800000	4.920000	30.200000	3355.000000
50%	19.300000	35.000000	6.320000	43.300000	9960.000000
75%	62.100000	51.350000	8.600000	58.750000	22800.000000
max	208.000000	200.000000	17.900000	174.000000	125000.000000

	inflation	life_expec	total_fer	gdpp
count	167.000000	167.000000	167.000000	167.000000
mean	7.781832	70.555689	2.947964	12964.155689
std	10.570704	8.893172	1.513848	18328.704809
min	-4.210000	32.100000	1.150000	231.000000
25%	1.810000	65.300000	1.795000	1330.000000
50%	5.390000	73.100000	2.410000	4660.000000
75%	10.750000	76.800000	3.880000	14050.000000
max	104.000000	82.800000	7.490000	105000.000000

```
[7]: df.dtypes
```

```
[7]: country      object
child_mort      float64
exports         float64
health          float64
imports         float64
income          int64
inflation       float64
life_expec      float64
total_fer       float64
gdpp            int64
dtype: object
```

```
[8]: def boxplots_variables_numericas_plotly(df):
    df_numerico = df.select_dtypes(include=['number']).
    ↪drop(columns=['CustomerID'], errors='ignore')

    if df_numerico.empty:
        print(" No hay columnas numéricas para graficar")
        return

    print("Columnas numéricas:", df_numerico.columns.tolist())

    df_long = df_numerico.melt(var_name='Variable', value_name='Valor')

    fig = px.box(df_long,
                  x='Variable',
                  y='Valor',
                  color='Variable',
```

```

        facet_col='Variable',
        facet_col_wrap=3,
        title='Boxplots de Variables Numéricas')

fig.update_xaxes(showticklabels=False)
fig.update_layout(height=700, width=1000)
fig.show()

```

```
[9]: boxplots_variables_numericas_plotly(df)
```

Columnas numéricas: ['child_mort', 'exports', 'health', 'imports', 'income', 'inflation', 'life_expec', 'total_fer', 'gdpp']

```
[10]: # existencia de NA (datos faltantes)
df.isna().sum()
```

```
[10]: country      0
      child_mort   0
      exports      0
      health       0
      imports      0
      income       0
      inflation    0
      life_expec   0
      total_fer    0
      gdpp         0
      dtype: int64
```

```
[11]: # Debemos trabajar con datos no nulos. En esta ocasión los vamos a omitir
df2 = df.dropna()
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 167 entries, 0 to 166
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   country         167 non-null   object
 1   child_mort      167 non-null   float64
 2   exports         167 non-null   float64
 3   health          167 non-null   float64
 4   imports         167 non-null   float64
 5   income          167 non-null   int64
 6   inflation       167 non-null   float64
 7   life_expec      167 non-null   float64
 8   total_fer       167 non-null   float64
 9   gdpp            167 non-null   int64
dtypes: float64(7), int64(2), object(1)

```

memory usage: 13.2+ KB

```
[12]: # vemos la cabeza de los datos
df2.head()
```

```
[12]:
```

	country	child_mort	exports	health	imports	income	\
0	Afghanistan	90.2	10.0	7.58	44.9	1610	
1	Albania	16.6	28.0	6.55	48.6	9930	
2	Algeria	27.3	38.4	4.17	31.4	12900	
3	Angola	119.0	62.3	2.85	42.9	5900	
4	Antigua and Barbuda	10.3	45.5	6.03	58.9	19100	

	inflation	life_expec	total_fer	gdpp
0	9.44	56.2	5.82	553
1	4.49	76.3	1.65	4090
2	16.10	76.5	2.89	4460
3	22.40	60.1	6.16	3530
4	1.44	76.8	2.13	12200

```
[13]: df2
```

```
[13]:
```

	country	child_mort	exports	health	imports	income	\
0	Afghanistan	90.2	10.0	7.58	44.9	1610	
1	Albania	16.6	28.0	6.55	48.6	9930	
2	Algeria	27.3	38.4	4.17	31.4	12900	
3	Angola	119.0	62.3	2.85	42.9	5900	
4	Antigua and Barbuda	10.3	45.5	6.03	58.9	19100	
..	...	...	...	...	...	...	
162	Vanuatu	29.2	46.6	5.25	52.7	2950	
163	Venezuela	17.1	28.5	4.91	17.6	16500	
164	Vietnam	23.3	72.0	6.84	80.2	4490	
165	Yemen	56.3	30.0	5.18	34.4	4480	
166	Zambia	83.1	37.0	5.89	30.9	3280	

	inflation	life_expec	total_fer	gdpp
0	9.44	56.2	5.82	553
1	4.49	76.3	1.65	4090
2	16.10	76.5	2.89	4460
3	22.40	60.1	6.16	3530
4	1.44	76.8	2.13	12200
..	...	...	...	...
162	2.62	63.0	3.50	2970
163	45.90	75.4	2.47	13500
164	12.10	73.1	1.95	1310
165	23.60	67.5	4.67	1310
166	14.00	52.0	5.40	1460

[167 rows x 10 columns]

2.3 1.3 Estandarización

```
[14]: # escalar los datos
```

```
df2_numeric = df2.select_dtypes(include=np.number)
df2_std = StandardScaler().fit_transform(df2_numeric)
df2_std
```

```
[14]: array([[ 1.29153238, -1.13827979,  0.27908825, ..., -1.61909203,
          1.90288227, -0.67917961],
        [-0.5389489 , -0.47965843, -0.09701618, ...,  0.64786643,
          -0.85997281, -0.48562324],
        [-0.27283273, -0.09912164, -0.96607302, ...,  0.67042323,
          -0.0384044 , -0.46537561],
        ...,
        [-0.37231541,  1.13030491,  0.0088773 , ...,  0.28695762,
          -0.66120626, -0.63775406],
        [ 0.44841668, -0.40647827, -0.59727159, ..., -0.34463279,
          1.14094382, -0.63775406],
        [ 1.11495062, -0.15034774, -0.33801514, ..., -2.09278484,
          1.6246091 , -0.62954556]])
```

```
[15]: # revisamos la matriz de correlaciones
```

```
pd.DataFrame(df2_std , index = df2.index , columns = df2_numeric.columns).corr()
```

```
[15]:
```

	child_mort	exports	health	imports	income	inflation \
child_mort	1.000000	-0.318093	-0.200402	-0.127211	-0.524315	0.288276
exports	-0.318093	1.000000	-0.114408	0.737381	0.516784	-0.107294
health	-0.200402	-0.114408	1.000000	0.095717	0.129579	-0.255376
imports	-0.127211	0.737381	0.095717	1.000000	0.122406	-0.246994
income	-0.524315	0.516784	0.129579	0.122406	1.000000	-0.147756
inflation	0.288276	-0.107294	-0.255376	-0.246994	-0.147756	1.000000
life_expec	-0.886676	0.316313	0.210692	0.054391	0.611962	-0.239705
total_fer	0.848478	-0.320011	-0.196674	-0.159048	-0.501840	0.316921
gdpp	-0.483032	0.418725	0.345966	0.115498	0.895571	-0.221631

	life_expec	total_fer	gdpp
child_mort	-0.886676	0.848478	-0.483032
exports	0.316313	-0.320011	0.418725
health	0.210692	-0.196674	0.345966
imports	0.054391	-0.159048	0.115498
income	0.611962	-0.501840	0.895571
inflation	-0.239705	0.316921	-0.221631
life_expec	1.000000	-0.760875	0.600089
total_fer	-0.760875	1.000000	-0.454910
gdpp	0.600089	-0.454910	1.000000

2.4 1.4 Análisis de PCA

```
[16]: # Definimos el objeto PCA y determinamos el número de componentes principales
      ↪ que deseamos.
pca = PCA(n_components= 0.85)                # Otra opción pca =
      ↪ PCA(n_components = 0.85).
pca.fit(df2_std)                            # Ajuste ese PCA a los datos
      ↪ estandarizados.
pca_proj = pca.transform(df2_std)           # calcula las nuevas variables que son
      ↪ combinación lineal de las originales
pca_proj
```

```
[16]: array([[ -2.91302459e+00,  9.56205755e-02, -7.18118495e-01,
               1.00525464e+00],
              [ 4.29911330e-01, -5.88155666e-01, -3.33485505e-01,
               -1.16105859e+00],
              [-2.85225077e-01, -4.55174413e-01,  1.22150481e+00,
               -8.68114503e-01],
              [-2.93242265e+00,  1.69555507e+00,  1.52504374e+00,
               8.39625014e-01],
              [ 1.03357587e+00,  1.36658709e-01, -2.25720917e-01,
               -8.47062687e-01],
              [ 2.24072616e-02, -1.77918658e+00,  8.69997116e-01,
               -3.69668667e-02],
              [-1.01583737e-01, -5.68251724e-01,  2.42091816e-01,
               -1.46626576e+00],
              [ 2.34216461e+00, -1.98845915e+00,  1.90344188e-01,
               1.10503778e+00],
              [ 2.97376366e+00, -7.34688659e-01, -5.19766356e-01,
               1.20544210e+00],
              [-1.81486997e-01, -4.02865873e-01,  8.67458743e-01,
               -4.38772983e-01],
              [ 1.26874386e+00, -6.56588363e-01, -4.88097616e-01,
               5.56335553e-02],
              [ 1.67099640e+00,  5.61162493e-01,  9.91258303e-01,
               -2.07080475e-01],
              [-1.12385093e+00, -9.61397405e-01,  5.26614801e-01,
               -1.19720139e+00],
              [ 1.08137420e+00, -4.81969530e-01, -6.64355198e-01,
               -5.22505233e-01],
              [ 5.80025152e-01,  5.35326834e-01,  4.86227511e-01,
               -1.03527542e+00],
              [ 3.14378596e+00,  6.63547921e-01, -6.45228094e-01,
               1.09489775e+00],
              [ 2.11255447e-01,  6.99242662e-01, -2.75560900e-01,
               -1.09562692e+00],
              [-2.67231388e+00,  4.18172125e-01, -2.57368473e-01,
```

2.78672434e-01],
 [-1.56570962e-01, 7.77395617e-01, -2.77343559e-01,
 -1.03865011e+00],
 [-7.93851561e-01, -1.20261085e-01, 4.00741495e-01,
 -7.24703290e-01],
 [9.95867143e-01, -9.71888439e-01, -1.52832760e+00,
 -6.20884792e-01],
 [-8.82087639e-01, 4.57368180e-01, -5.84632949e-01,
 4.06160810e-01],
 [1.40781361e-01, -2.15107731e+00, -1.72635153e-01,
 -1.66919647e-01],
 [2.46008609e+00, 1.64540436e-02, 3.01984039e+00,
 7.76840751e-01],
 [9.06594515e-01, 3.02776054e-02, -4.71204036e-01,
 -9.94661396e-01],
 [-3.12205344e+00, 3.87749688e-02, -4.55750797e-01,
 1.08091762e+00],
 [-2.89897068e+00, -4.22663328e-01, -1.34741345e+00,
 1.82147410e+00],
 [-5.82411867e-01, 8.94820332e-01, -4.98704660e-01,
 -8.05233622e-01],
 [-2.80790857e+00, 7.86488969e-02, -3.42961116e-01,
 5.43556844e-01],
 [2.54363055e+00, -1.72709470e+00, -3.96281023e-01,
 1.33685255e+00],
 [-1.55801452e-01, 3.51235458e-01, -2.92086671e-01,
 -1.34374110e+00],
 [-3.96496402e+00, 3.86619319e-01, -3.23556922e-01,
 9.18301019e-01],
 [-3.55755520e+00, 1.28912809e+00, -1.03015433e-01,
 1.20840642e+00],
 [9.51656055e-01, -1.07642827e+00, 1.50820820e-01,
 -4.36634193e-01],
 [5.74819803e-02, -1.18999652e+00, 4.27315488e-01,
 -1.27717888e+00],
 [1.21146120e-01, -1.76890914e+00, -2.45605887e-01,
 -7.15208871e-01],
 [-2.09355643e+00, 3.43600988e-01, -3.28381272e-01,
 -1.18947248e-01],
 [-3.17337012e+00, 1.05038163e+00, -1.21915456e-03,
 1.57421016e+00],
 [-1.72567641e+00, 2.17634895e+00, 1.49271940e+00,
 -1.38592628e-01],
 [9.37826615e-01, -1.35047238e+00, -8.21129661e-01,
 -2.59854592e-01],
 [-2.58170623e+00, 1.20787342e+00, -2.33066157e-01,
 6.65293359e-01],

[1.14886344e+00, -8.44812046e-01, -4.09712564e-01,
 -6.22922553e-01],
 [2.17445492e+00, -4.51044737e-03, 2.57319634e-01,
 -3.11856544e-01],
 [2.05326329e+00, 4.23198280e-01, -5.43335682e-01,
 -4.11735959e-01],
 [3.01049182e+00, -8.65548729e-01, -3.35752167e-01,
 1.78722278e+00],
 [-2.31102923e-01, -8.80641302e-01, -3.67363816e-02,
 -6.53432853e-01],
 [9.61833240e-03, -1.04522097e+00, -2.73292441e-01,
 -4.88513719e-01],
 [-8.48186699e-01, -8.19818902e-01, 5.83289752e-01,
 -7.45230918e-01],
 [8.18678445e-02, -5.67803943e-01, -5.75877586e-01,
 -9.76520622e-01],
 [-1.29342284e+00, 2.36369455e+00, 1.77933449e+00,
 1.73724419e+00],
 [-2.47469590e+00, -6.18025236e-01, 8.03496573e-01,
 -4.88256238e-01],
 [1.65908340e+00, 1.02156447e+00, -1.11116210e-01,
 -8.63149640e-01],
 [-1.88828409e-01, 1.07176458e+00, -1.90494207e-01,
 -9.79835597e-01],
 [2.45896019e+00, -1.07614294e+00, -1.00439032e-01,
 8.99328358e-01],
 [2.25427080e+00, -1.86663813e+00, -7.11988312e-01,
 1.23000548e+00],
 [-1.42171455e+00, 3.19723358e-01, 1.53753336e+00,
 1.87759190e-01],
 [-2.21366958e+00, 2.23495896e-01, -3.93111292e-01,
 2.89914182e-01],
 [3.21942207e-01, -5.18255225e-01, -9.73687319e-01,
 -4.67061703e-01],
 [2.67142195e+00, -1.27360990e+00, -6.56170834e-01,
 1.12599297e+00],
 [-2.05416693e+00, 3.80034393e-01, 4.19654109e-01,
 9.09709651e-02],
 [1.77949294e+00, -1.76539693e+00, -6.84197774e-01,
 2.62701918e-01],
 [1.45504799e-01, -4.31336366e-01, -4.70467998e-01,
 -9.71167871e-01],
 [-6.63503125e-01, -6.13910837e-01, -3.32673741e-01,
 -4.45317091e-01],
 [-2.96952947e+00, 7.28533786e-01, 3.91038978e-01,
 6.66549494e-01],
 [-2.83361647e+00, -9.11281950e-02, -1.20517398e+00,

1.13304489e+00],
 [-3.22781465e-01, 1.36134136e+00, -4.70512158e-01,
 -8.50455474e-01],
 [-4.40971727e+00, 1.74223049e+00, -1.46280874e+00,
 1.79359222e+00],
 [1.83916013e+00, 1.27296493e+00, -4.62631365e-01,
 -7.79275693e-01],
 [2.48092396e+00, -6.34701926e-01, 1.03795226e-01,
 9.19853150e-01],
 [-1.34282579e+00, -5.35138946e-01, 4.69946406e-01,
 -7.88293802e-01],
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 -1.14368788e+00],
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 5.16711898e-01],
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 7.12566269e-01],
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 -4.53078232e-01],
 [-1.87470247e+00, -1.71029967e-01, -2.49557050e-01,
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 -4.15581903e-01],
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 -9.30854536e-01],
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 -1.05418607e+00],
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 1.07219875e+00],

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 1.17956908e+00],
 [8.18943051e-01, 3.89841660e-01, 1.48989162e+00,
 -6.19545661e-01],
 [1.40978812e+00, 7.29833198e-01, -4.08006476e-01,
 -7.64677732e-01],
 [6.91775496e+00, 4.84984369e+00, 1.00183722e+00,
 3.12964255e+00],
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 -1.13823279e+00],
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 -3.17468942e-01],
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 8.39108028e-01],
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 -1.08637235e+00],
 [1.10860101e+00, 1.00931426e+00, -2.69979852e-01,
 -1.08907107e+00],
 [-3.41225513e+00, 5.61468514e-01, -2.48615098e-01,
 1.07336391e+00],
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 -5.01107355e-01],
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 1.95412788e-01],
 [8.99775055e-01, 4.16479781e-01, -3.81759043e-01,
 -1.07218433e+00],
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 7.71973760e-01],
 [5.09539453e-01, 1.61658340e-01, -1.56585938e+00,
 -4.18362236e-01],
 [-9.44975538e-01, 5.29799562e-01, 1.85441886e+00,
 -3.94165588e-01],
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 -7.15644105e-01],
 [-2.32870156e-01, -2.81027769e-01, -2.61356785e-01,
 -1.07639425e+00],
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7.35652166e-01],
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 2.41242140e+00],
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 2.76433853e+00],
 [1.12738665e+00, 4.91611136e-01, 2.13350091e+00,
 -1.76492480e-01],
 [-2.36034718e+00, -4.79399646e-01, 1.01355540e+00,
 -4.21709938e-01],
 [1.16378429e+00, 1.11527620e+00, -7.90119264e-01,
 -5.45194408e-01],
 [1.17846224e-01, 3.61031140e-01, -4.66723186e-02,
 -9.46838629e-01],
 [-2.06354519e-02, -1.08661741e+00, 4.04348774e-01,
 -1.06962172e+00],
 [-7.82745871e-01, -9.64980905e-02, 3.09094042e-01,
 -1.08235769e+00],
 [1.21782754e+00, -6.59168961e-01, -3.11232316e-01,
 -6.95175223e-01],
 [1.81406748e+00, -1.45088654e+00, -9.42487215e-01,
 1.93791564e-01],
 [4.24229634e+00, -1.95603674e-01, 3.64387511e+00,
 2.38981058e+00],
 [5.72792704e-01, -6.37384843e-01, 8.63507590e-02,
 -1.02983930e+00],
 [1.63761544e-01, -1.06667848e+00, 1.09051847e+00,
 -6.44874077e-01],
 [-1.67970356e+00, -1.00162862e+00, -1.46843591e+00,
 6.73220742e-01],
 [-5.62897632e-01, -2.21043960e-02, -6.38390549e-01,
 -5.32651338e-01],
 [8.55935813e-01, -1.83440759e-01, 1.90307225e+00,
 2.02258954e-01],
 [-1.91217031e+00, 9.15599347e-02, -5.06958976e-01,
 2.76252708e-02],
 [8.32420187e-01, -8.69325996e-01, -1.01769287e+00,
 -5.03134235e-01],
 [1.60259775e+00, 2.93912057e+00, -3.86691417e-01,
 -1.40037119e+00],
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 2.58197644e+00],
 [5.78337630e+00, 6.68209028e+00, 6.54420074e-01,
 8.96376420e-02],
 [2.02972370e+00, 1.05040745e+00, -8.54978429e-01,
 -3.71447784e-01],

[2.27949171e+00, 1.95275226e-01, -8.01535444e-01,
 -6.73604956e-02],
 [-8.06209136e-01, 1.30349059e+00, -1.22922699e+00,
 -4.23618753e-02],
 [-1.19183736e+00, -5.56757164e-01, -7.17965953e-01,
 5.50205983e-01],
 [1.91806245e+00, -4.27468245e-01, 1.40741993e-01,
 -4.72252955e-01],
 [2.01919721e+00, -1.78438246e+00, -3.80316498e-01,
 2.64389054e-01],
 [-5.75572155e-01, -9.97551478e-01, 1.75224072e+00,
 -1.37803039e+00],
 [2.66234652e-02, -1.60640815e-02, -3.85463343e-02,
 -1.22484232e+00],
 [-2.31942387e+00, -7.69407328e-01, 7.32740073e-01,
 4.92083043e-01],
 [1.71674731e-01, -9.48076409e-02, 1.20264550e-02,
 -3.76833834e-01],
 [2.81832286e+00, -9.14480968e-01, -1.01807170e-01,
 1.24227931e+00],
 [4.08854413e+00, -4.29461909e-01, -2.61547109e-01,
 2.30708671e+00],
 [-1.24446436e+00, -2.89174316e-02, -1.48159392e-01,
 -4.81045924e-01],
 [-2.55404919e+00, -2.15027956e-01, -8.84904657e-02,
 4.62226441e-01],
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 -1.59519938e+00],
 [-2.37197047e+00, -1.17751295e+00, 3.98922832e-01,
 1.15804501e+00],
 [-1.99764225e+00, 9.58361586e-01, -1.21803661e+00,
 5.60916606e-01],
 [-7.55008538e-01, -8.78938568e-02, -4.23287693e-01,
 -8.54637175e-01],
 [6.02231612e-01, 1.73435708e-01, -2.43158206e-01,
 -1.11793721e+00],
 [4.01437705e-01, -1.41198973e+00, 2.25685264e-01,
 -5.63294782e-01],
 [-4.63936165e-01, 1.29187347e+00, 5.99238344e-01,
 -9.39834172e-01],
 [-2.85483624e+00, -3.52082382e-01, -7.00859396e-01,
 1.29627050e+00],
 [3.02299800e-01, -9.75710669e-02, -9.06375587e-02,
 -8.79675120e-01],
 [2.42714125e+00, 1.15181307e+00, 1.82853811e+00,
 2.76080223e-01],
 [2.06798993e+00, -1.53531349e+00, -2.52772015e-01,

```

7.91524403e-01],
[ 2.64120583e+00, -2.99736446e+00, -1.72187786e+00,
 2.82711027e+00],
[ 6.17312598e-01, -1.43047723e+00, -2.44436640e-01,
-3.58717593e-01],
[-8.53528944e-01, -6.54485112e-01,  6.19884667e-01,
-7.22182872e-01],
[-8.20631131e-01,  6.39570072e-01, -3.89923152e-01,
-7.06595241e-01],
[-5.51035564e-01, -1.23388618e+00,  3.10135006e+00,
-1.15311025e-01],
[ 4.98524385e-01,  1.39074432e+00, -2.38526113e-01,
-1.07409791e+00],
[-1.88745106e+00, -1.09453015e-01,  1.10975159e+00,
 5.62567749e-02],
[-2.86406392e+00,  4.85997985e-01,  2.23166581e-01,
 8.16364358e-01]])

```

```
[17]: pca.components_
```

```

[17]: array([[ -0.41951945,  0.28389698,  0.15083782,  0.16148244,  0.39844111,
        -0.19317293,  0.42583938, -0.40372896,  0.39264482],
       [ 0.19288394,  0.61316349, -0.24308678,  0.67182064,  0.02253553,
        -0.00840447, -0.22270674,  0.15523311, -0.0460224 ],
       [-0.02954353,  0.14476069, -0.59663237, -0.29992674,  0.3015475 ,
        0.64251951,  0.11391854,  0.01954925,  0.12297749],
       [ 0.37065326,  0.00309102,  0.4618975 , -0.07190746,  0.39215904,
        0.15044176, -0.20379723,  0.37830365,  0.53199457]])

```

```

[18]: proyecciones = pd.DataFrame(pca_proj,
                                columns = [f'PC{i+1}' for i in range(pca_proj.
                                ↪shape[1])],
                                index   = df2.index)
proyecciones.head(10)

```

```

[18]:
   PC1    PC2    PC3    PC4
0 -2.913025  0.095621 -0.718118  1.005255
1  0.429911 -0.588156 -0.333486 -1.161059
2 -0.285225 -0.455174  1.221505 -0.868115
3 -2.932423  1.695555  1.525044  0.839625
4  1.033576  0.136659 -0.225721 -0.847063
5  0.022407 -1.779187  0.869997 -0.036967
6 -0.101584 -0.568252  0.242092 -1.466266
7  2.342165 -1.988459  0.190344  1.105038
8  2.973764 -0.734689 -0.519766  1.205442
9 -0.181487 -0.402866  0.867459 -0.438773

```

```
[19]: proyecciones.corr()
```

```
[19]:
```

	PC1	PC2	PC3	PC4
PC1	1.000000e+00	1.109381e-16	-3.233265e-17	3.670921e-17
PC2	1.109381e-16	1.000000e+00	4.336341e-17	2.706869e-16
PC3	-3.233265e-17	4.336341e-17	1.000000e+00	2.663178e-16
PC4	3.670921e-17	2.706869e-16	2.663178e-16	1.000000e+00

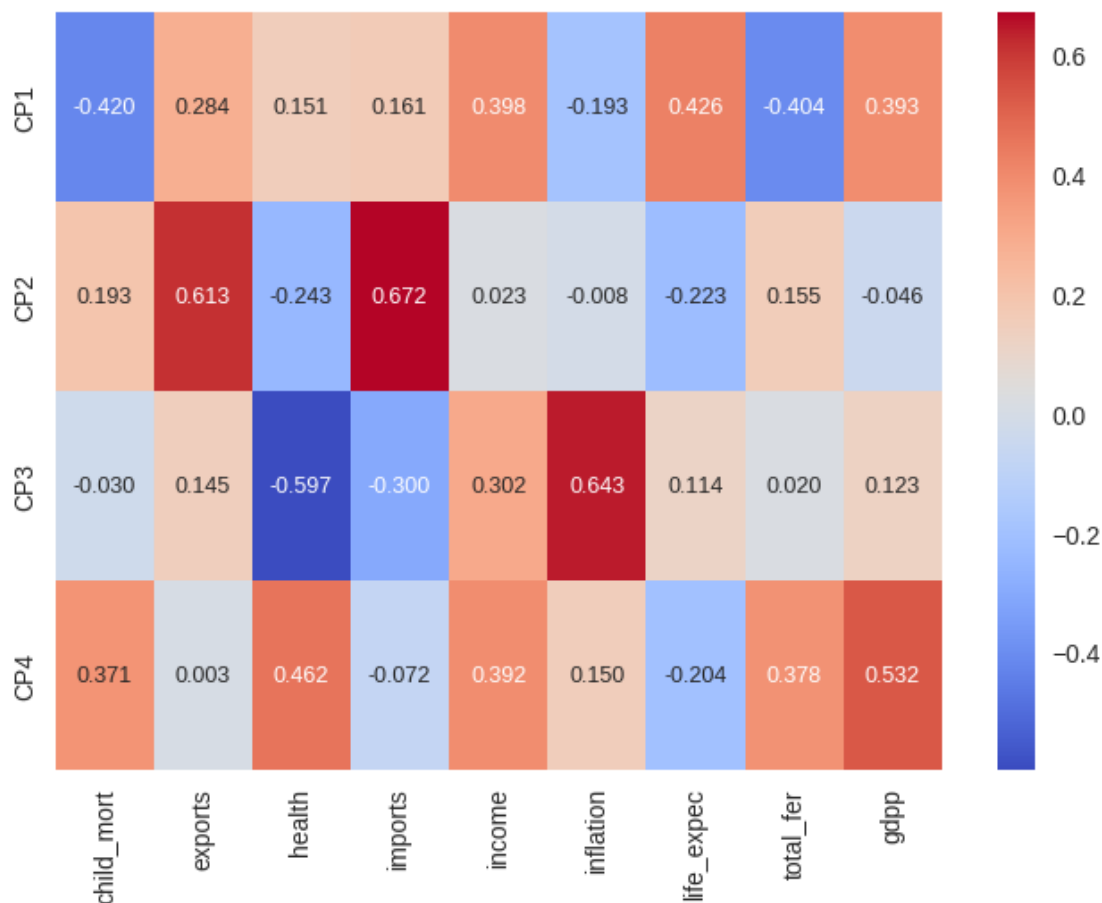
```
[20]: # para ver la varianza explicada
pca.explained_variance_ratio_
```

```
[20]: array([0.4595174 , 0.17181626, 0.13004259, 0.11053162])
```

```
[21]: # Cantidad de varianza retenida por las 2 primeras componentes principales.
ratio = np.sum(pca.explained_variance_ratio_)
print(f"\nLa cantidad de variabilidad retenida por las 2 primeras componentes_
      es {ratio*100:.3f}%")
```

La cantidad de variabilidad retenida por las 2 primeras componentes es 87.191%

```
[22]: sns.heatmap(pca.components_,
                  xticklabels=df2_numeric.columns,
                  yticklabels=["CP1", "CP2", "CP3", "CP4"],
                  cmap="coolwarm",
                  fmt=".3f",
                  annot=True,
                  annot_kws={"size": 9})
plt.figure(figsize=(12, 5))
plt.tight_layout()
plt.show()
```



<Figure size 1200x500 with 0 Axes>

2.5 ANÁLISIS

CP1 captura el eje de desarrollo humano: carga negativa fuerte en `life_expec` y positiva en `child_mort` y `total_fer`, separando países desarrollados de subdesarrollados. CP2 está dominada por `exports` e `imports`, representando apertura comercial. CP3 refleja principalmente la `inflation`. CP4 integra `income` y `gdpp`, capturando prosperidad económica absoluta. En conjunto las 4 componentes explican el 87.19% de la varianza total.

2.6 1.5 Varianza explicada acumulada

```
[23]: # Definimos el objeto PCA y determinamos el número de componentes principales
      ↪ que deseamos.
pca3 = PCA(n_components=3)                                # Otra opción pca =
      ↪ PCA(n_components = 0.85).
pca3.fit(df2_std)                                         # Ajuste ese PCA a los datos
      ↪ estandarizados.
```

```
pca3_proj = pca3.transform(df2_std)          # calcula las nuevas variables que
↳ son combinación lineal de las originales
```

```
[24]: proyecciones3 = pd.DataFrame(pca3.transform(df2_std),
                                   columns = ["PC1", "PC2", "PC3"],
                                   index   = df2.index
                                   )
proyecciones3.head(10)
```

```
[24]:      PC1      PC2      PC3
0 -2.913025  0.095621 -0.718118
1  0.429911 -0.588156 -0.333486
2 -0.285225 -0.455174  1.221505
3 -2.932423  1.695555  1.525044
4  1.033576  0.136659 -0.225721
5  0.022407 -1.779187  0.869997
6 -0.101584 -0.568252  0.242092
7  2.342165 -1.988459  0.190344
8  2.973764 -0.734689 -0.519766
9 -0.181487 -0.402866  0.867459
```

```
[25]: # Porcentaje de varianza explicada acumulada
# =====
prop_varianza_acum = pca3.explained_variance_ratio_.cumsum()
print('-----')
print('Porcentaje de varianza explicada acumulada')
print('-----')
print(prop_varianza_acum)

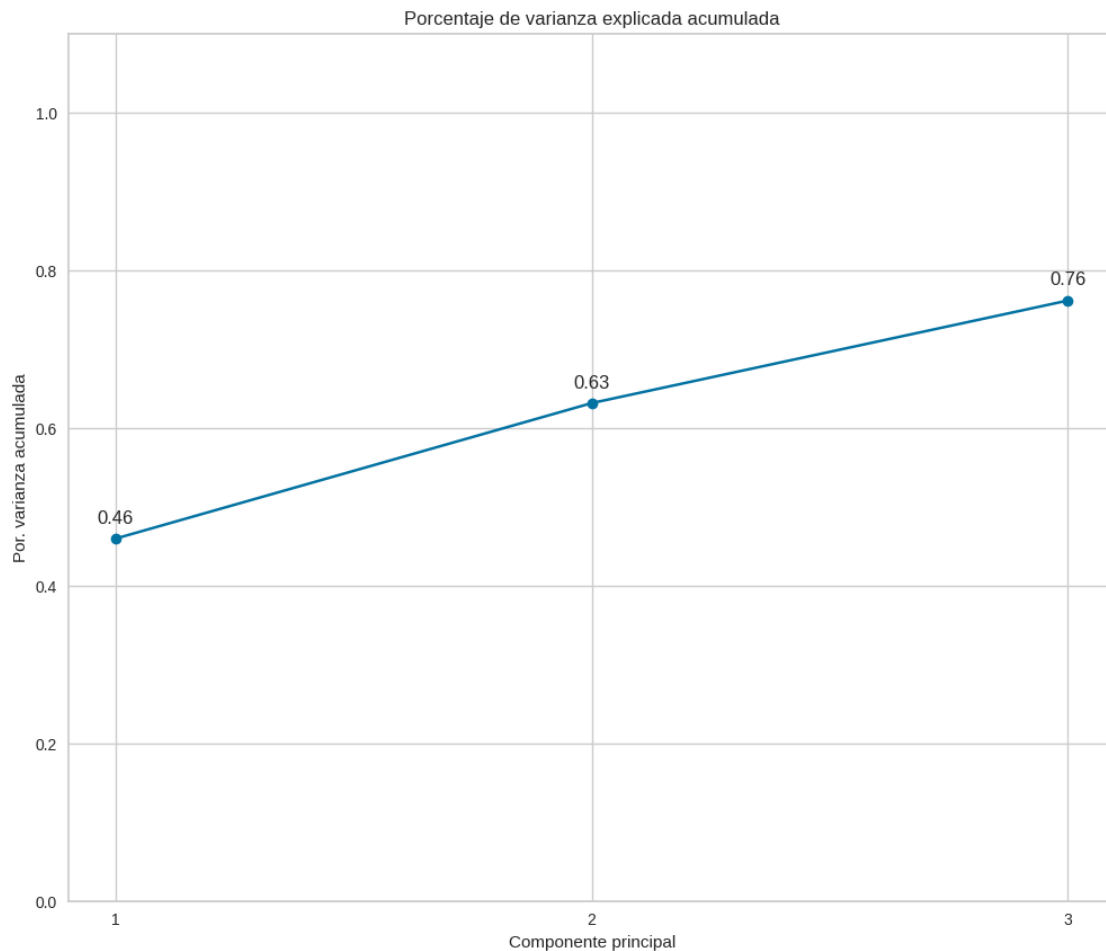
fig, ax = plt.subplots(nrows=1, ncols=1, figsize=(12, 10))
ax.plot(
    np.arange(len(proyecciones3.columns)) + 1,
    prop_varianza_acum,
    marker = 'o'
)

for x, y in zip(np.arange(len(proyecciones3.columns)) + 1, prop_varianza_acum):
    ↳ # hacer las proyecciones
    label = round(y, 2)
    ax.annotate(
        label,
        (x,y),
        textcoords="offset points",
        xytext=(0,10),
        ha='center'
    )
```

```
ax.set_ylim(0, 1.1)
ax.set_xticks(np.arange(pca3.n_components_) + 1)
ax.set_title('Porcentaje de varianza explicada acumulada')
ax.set_xlabel('Componente principal')
ax.set_ylabel('Por. varianza acumulada');
```

Porcentaje de varianza explicada acumulada

[0.4595174 0.63133365 0.76137624]



2.7 ANÁLISIS

Con solo 3 componentes se retiene el 76.1% de la información original, y con las 4 componentes que seleccionó automáticamente el PCA (`n_components=0.85`) se alcanza el 87.2%. Esto confirma que la estructura del dataset puede resumirse eficientemente en pocas dimensiones sin pérdida significativa de información.

2.8 1.6 Proyección de países

```
[26]: fig = px.scatter_3d(proyecciones3,
                        x="PC1",
                        y="PC2",
                        z="PC3",
                        hover_name=df2.index,
                        text=df2.index,
                        labels={'0': 'PC 1', '1': 'PC 2', '2': 'PC 3'})

fig.show()
```

```
[27]: fig=px.scatter(proyecciones,
                    x="PC1",
                    y="PC2",
                    hover_name=proyecciones.index,
                    text=proyecciones.index)

fig.show()
```

2.9 ANÁLISIS

Los países se distribuyen a lo largo de PC1 principalmente, confirmando que el eje de desarrollo humano es la mayor fuente de variación. Se identifican outliers claros como los índices 133, 91 y 98, que corresponden a países con perfiles extremos en mortalidad infantil, exportaciones o ingreso. La dispersión horizontal (PC1) es mayor que la vertical (PC2), consistente con que PC1 explica mayor proporción de varianza.

2.10 1.7 Biplot

```
[28]: !pip install pca
      #bipots
```

Collecting pca

Downloading pca-2.10.2-py3-none-any.whl.metadata (12 kB)

Collecting datazets (from pca)

Downloading datazets-1.1.3-py3-none-any.whl.metadata (13 kB)

Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-packages (from pca) (0.14.6)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from pca) (3.10.0)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from pca) (2.0.2)

Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from pca) (1.6.1)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from pca) (1.16.3)

Collecting colourmap>=1.1.19 (from pca)

Downloading colourmap-1.2.1-py3-none-any.whl.metadata (4.3 kB)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages

```

(from pca) (2.2.2)
Collecting scatterd>=1.3.7 (from pca)
  Downloading scatterd-1.4.2-py3-none-any.whl.metadata (5.6 kB)
Collecting adjusttext (from pca)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (from scatterd>=1.3.7->pca) (0.13.2)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from datazets->pca) (2.32.4)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (1.3.3)
Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->pca) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->pca) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->pca) (2026.1)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->pca) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->pca) (3.6.0)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from statsmodels->pca) (1.0.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->pca) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->datazets->pca) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->datazets->pca) (3.13)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->datazets->pca) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->datazets->pca) (2026.4.22)
Downloading pca-2.10.2-py3-none-any.whl (35 kB)
Downloading colourmap-1.2.1-py3-none-any.whl (10 kB)

```

Downloading scatterd-1.4.2-py3-none-any.whl (13 kB)
Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)
Downloading datazets-1.1.3-py3-none-any.whl (190 kB)
190.4/190.4 kB

7.1 MB/s eta 0:00:00

Installing collected packages: datazets, colourmap, adjusttext, scatterd,
pca
Successfully installed adjusttext-1.3.0 colourmap-1.2.1 datazets-1.1.3
pca-2.10.2 scatterd-1.4.2

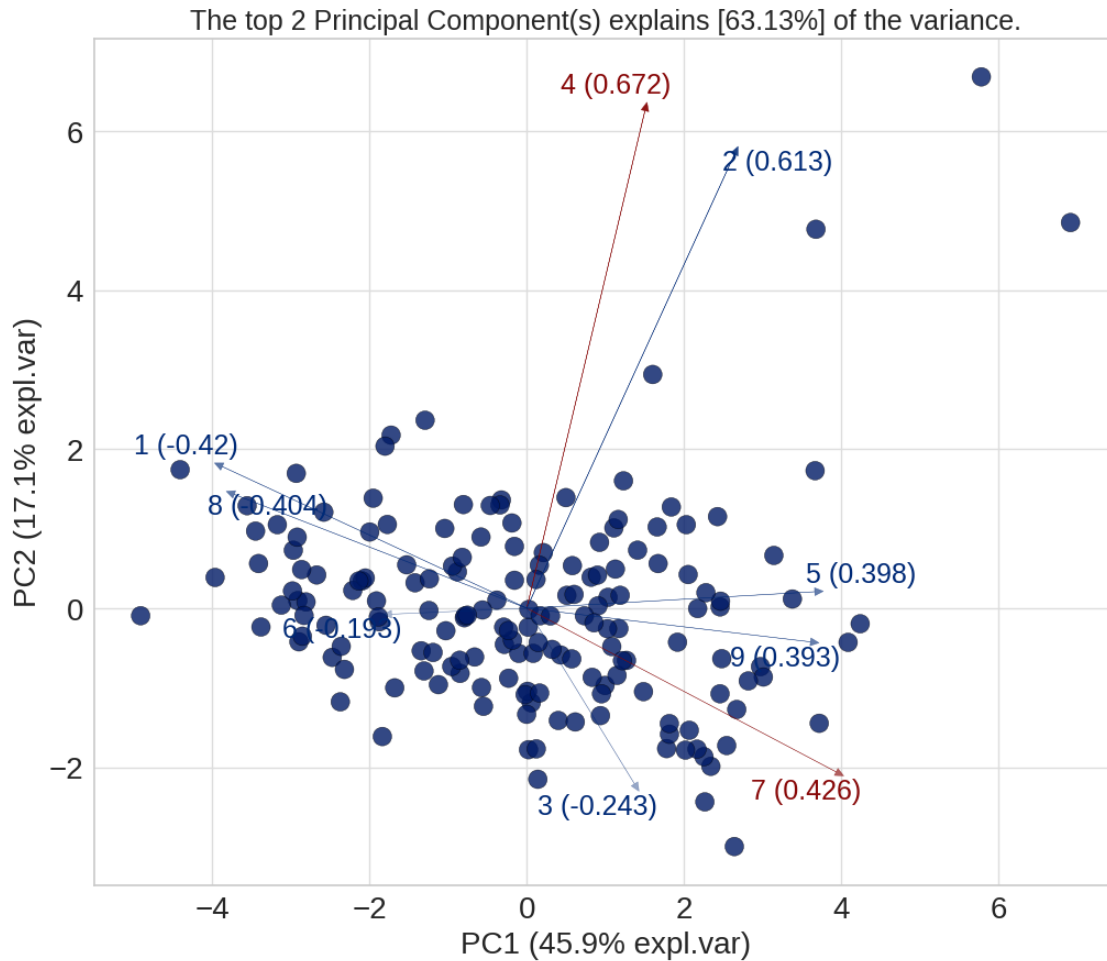
```
[29]: from pca import pca
```

```
[30]: model = pca(n_components=2, normalize=True)  
out = model.fit_transform(df2_std)
```

```
[09-05-2026 17:11:55] [pca.pca] [INFO] Column labels are auto-completed.  
[09-05-2026 17:11:55] [pca.pca] [INFO] Row labels are auto-completed.  
[09-05-2026 17:11:55] [pca.pca] [INFO] Normalizing input data per feature (zero  
mean and unit variance)..  
[09-05-2026 17:11:55] [pca.pca] [INFO] The PCA reduction is performed on 9  
variables (columns) of the input dataframe.  
[09-05-2026 17:11:55] [pca.pca] [INFO] Fit using PCA.  
[09-05-2026 17:11:55] [pca.pca] [INFO] Compute loadings and PCs.  
[09-05-2026 17:11:55] [pca.pca] [INFO] Compute explained variance.  
[09-05-2026 17:11:55] [pca.pca] [INFO] Outlier detection using Hotelling T2 test  
with alpha=[0.05] and n_components=[2]  
[09-05-2026 17:11:55] [pca.pca] [INFO] Multiple test correction applied for  
Hotelling T2 test: [fdr_bh]  
[09-05-2026 17:11:55] [pca.pca] [INFO] Outlier detection using SPE/DmodX with  
n_std=[3]
```

```
[31]: # para hacer el biplot  
fig, ax = model.biplot(label=True,  
                        legend=True,  
                        figsize=(12, 10))
```

```
[09-05-2026 17:11:55] [pca.pca] [WARNING] Parameter <label> is deprecated and  
will not be supported in future version.  
[09-05-2026 17:11:55] [pca.pca] [INFO] Plot PC1 vs PC2 with loadings.  
[09-05-2026 17:11:55] [scatterd.scatterd] [INFO] Create scatterplot
```



2.11 ANÁLISIS

Los vectores de `life_expec` y `child_mort` apuntan en direcciones opuestas sobre PC1, confirmando su correlación negativa. `exports` e `imports` se alinean positivamente con PC2. Los países en el cuadrante superior derecho combinan alta esperanza de vida con alta apertura comercial. Los vectores más largos (`child_mort`, `life_expec`, `exports`) son los de mayor capacidad explicativa dentro del plano.

2.12 Conclusiones

En conclusión, el Análisis de Componentes Principales permitió reducir eficazmente la complejidad del conjunto de datos, logrando que los dos primeros componentes expliquen el 63.13% de la variabilidad total y que tres componentes superen el 76%, lo cual demuestra que gran parte de la información original puede representarse mediante un número reducido de dimensiones. Los resultados evidencian que las principales diferencias entre los países analizados están determinadas por factores económicos, sociales, sanitarios y demográficos, donde variables como PIB, ingreso, salud, mortalidad infantil, inflación, fertilidad y esperanza de vida desempeñan un papel fundamental. El primer componente refleja principalmente contrastes asociados al desarrollo económico y estabilidad

macroeconómica, mientras que el segundo componente se relaciona con bienestar social y condiciones de salud pública. Asimismo, la proyección de las observaciones permitió identificar tanto grupos con características similares como casos atípicos con perfiles significativamente diferenciados, facilitando la detección de países con comportamientos excepcionales. En términos generales, el PCA demostró ser una herramienta altamente útil para simplificar, visualizar e interpretar estructuras complejas en datos multidimensionales, permitiendo comprender de manera más clara los patrones subyacentes que explican las desigualdades y relaciones entre variables socioeconómicas a nivel global.

3 2. K-means

Con los datos del punto anterior aplicar el método de k-means para hacer clusterización. Utilice el método del codo y silueta para determinar un número de clusters apropiado.

3.1 2.1 Librerías

```
[32]: import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots

from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import silhouette_score, silhouette_samples
from sklearn.cluster import KMeans
from sklearn.neighbors import NearestNeighbors
from sklearn.metrics import pairwise_distances_argmin_min

from yellowbrick.cluster import KElbowVisualizer

import warnings
warnings.filterwarnings("ignore")
```

3.2 2.2 Preprocesamiento

```
[33]: #solo variables numericas
df_num = df.select_dtypes(include=['number'])
# Escalamos los datos
scaler = StandardScaler()
scaler.fit(df_num)
df = pd.DataFrame(scaler.transform(df_num),
                  columns=df_num.columns)
df.head(6)
```

```
[33]: child_mort  exports  health  imports  income  inflation  life_expec  \
0    1.291532 -1.138280  0.279088 -0.082455 -0.808245  0.157336 -1.619092
1   -0.538949 -0.479658 -0.097016  0.070837 -0.375369 -0.312347  0.647866
2   -0.272833 -0.099122 -0.966073 -0.641762 -0.220844  0.789274  0.670423
3    2.007808  0.775381 -1.448071 -0.165315 -0.585043  1.387054 -1.179234
4   -0.695634  0.160668 -0.286894  0.497568  0.101732 -0.601749  0.704258
5   -0.591177 -0.812628  0.468966 -1.279787  0.080920  1.244725  0.591474

total_fer  gdpp
0    1.902882 -0.679180
1   -0.859973 -0.485623
2   -0.038404 -0.465376
3    2.128151 -0.516268
4   -0.541946 -0.041817
5   -0.382933 -0.145791
```

3.3 2.3 Modelo inicial

```
[34]: # generación del modelo con dos clusters
n = 9
kmeans = KMeans(n_clusters=n,                #Número de clusters
                 random_state=0)             #Semilla aleatoria.
kmeans.fit(df)
```

```
[34]: KMeans(n_clusters=9, random_state=0)
```

```
[35]: # Inercia y centroides
print(kmeans.inertia_)
print(kmeans.cluster_centers_)
```

```
464.3921228605916
[[-8.49003244e-01  4.93567278e+00 -8.16303241e-03  4.54805768e+00
  2.43954240e+00 -5.04206141e-01  1.22682431e+00 -1.03886271e+00
  2.44079735e+00]
 [-7.41437184e-02 -2.70256377e-02 -3.59788893e-01  2.48833005e-01
 -5.10488646e-01 -2.23221781e-01 -2.42709473e-01  5.26355932e-02
 -5.08051689e-01]
 [ 1.47603348e+00 -4.65975664e-01 -3.35420651e-01 -3.33979733e-01
 -7.21011531e-01  2.73283925e-01 -1.27272642e+00  1.54440505e+00
 -6.22208315e-01]
 [-8.28309571e-01  9.05838308e-02  1.00952855e+00 -4.09912691e-01
  1.52310836e+00 -4.99042175e-01  1.13153629e+00 -7.35871133e-01
  1.79246636e+00]
 [-3.29777991e-01 -4.49517036e-01 -5.52949942e-01 -7.96817047e-01
 -1.30440593e-01  3.41937785e-01  3.10292243e-01 -4.38450642e-01
 -3.32118670e-01]
 [-4.14595550e-01 -1.28393699e-01 -5.99097339e-01 -4.03538818e-01
```

```

-2.62207068e-01  3.29899826e+00  2.75544151e-02 -2.60360380e-01
-2.67550817e-01]
[-6.96877649e-01  4.82660565e-01  1.59907367e-01  5.49585392e-01
 8.71349099e-02 -3.85717755e-01  5.66724606e-01 -8.31446133e-01
-7.38910783e-02]
[ 2.28138502e+00 -5.78451631e-01 -6.37438082e-01 -1.22178479e+00
-6.24064774e-01  9.12971806e+00 -1.13412081e+00  1.91613337e+00
-5.81936274e-01]
[ 6.38926010e-01 -4.45995557e-01  1.67030949e+00  1.66755567e+00
-7.88682715e-01 -3.25631452e-01 -1.30104114e+00  6.78480305e-01
-6.31329540e-01]]

```

```

[36]: # Centroides como DataFrame
centroides = pd.DataFrame(kmeans.cluster_centers_, columns=df.columns)
centroides

```

```

[36]:   child_mort  exports  health  imports  income  inflation  life_expec \
0   -0.849003  4.935673 -0.008163  4.548058  2.439542  -0.504206    1.226824
1   -0.074144 -0.027026 -0.359789  0.248833 -0.510489  -0.223222   -0.242709
2    1.476033 -0.465976 -0.335421 -0.333980 -0.721012   0.273284   -1.272726
3   -0.828310  0.090584  1.009529 -0.409913  1.523108  -0.499042    1.131536
4   -0.329778 -0.449517 -0.552950 -0.796817 -0.130441   0.341938    0.310292
5   -0.414596 -0.128394 -0.599097 -0.403539 -0.262207   3.298998    0.027554
6   -0.696878  0.482661  0.159907  0.549585  0.087135  -0.385718    0.566725
7    2.281385 -0.578452 -0.637438 -1.221785 -0.624065   9.129718   -1.134121
8    0.638926 -0.445996  1.670309  1.667556 -0.788683  -0.325631   -1.301041

   total_fer  gdpp
0  -1.038863  2.440797
1   0.052636 -0.508052
2   1.544405 -0.622208
3  -0.735871  1.792466
4  -0.438451 -0.332119
5  -0.260360 -0.267551
6  -0.831446 -0.073891
7   1.916133 -0.581936
8   0.678480 -0.631330

```

```

[37]: # Atributos.
print(kmeans.labels_)

```

```

[2 6 4 2 6 4 4 3 3 4 6 6 4 6 6 3 1 2 1 1 6 1 4 3 6 2 2 1 2 3 1 2 2 4 4 4 2
 2 2 6 2 6 6 6 3 4 4 4 1 2 2 6 1 3 3 2 2 6 3 2 3 1 1 2 2 1 2 6 3 4 4 4 1 3
 3 3 4 3 1 4 2 8 3 1 1 6 6 8 8 4 6 0 6 2 2 6 6 2 0 2 6 8 6 5 6 1 2 4 1 4 3
 3 2 7 3 4 2 6 1 4 1 6 3 3 4 4 2 1 4 2 6 6 2 0 6 6 8 1 6 3 4 1 2 1 3 3 1 2
 6 2 2 1 6 4 1 2 6 6 3 3 4 4 1 5 6 2 2]

```

```
[38]: # Añadimos las etiquetas al DataFrame original
df_cluster = df_num.copy()
df_cluster['cluster'] = kmeans.labels_
df_cluster.head(10)
```

```
[38]:
```

	child_mort	exports	health	imports	income	inflation	life_expec	\
0	90.2	10.0	7.58	44.9	1610	9.440	56.2	
1	16.6	28.0	6.55	48.6	9930	4.490	76.3	
2	27.3	38.4	4.17	31.4	12900	16.100	76.5	
3	119.0	62.3	2.85	42.9	5900	22.400	60.1	
4	10.3	45.5	6.03	58.9	19100	1.440	76.8	
5	14.5	18.9	8.10	16.0	18700	20.900	75.8	
6	18.1	20.8	4.40	45.3	6700	7.770	73.3	
7	4.8	19.8	8.73	20.9	41400	1.160	82.0	
8	4.3	51.3	11.00	47.8	43200	0.873	80.5	
9	39.2	54.3	5.88	20.7	16000	13.800	69.1	

	total_fer	gdpp	cluster
0	5.82	553	2
1	1.65	4090	6
2	2.89	4460	4
3	6.16	3530	2
4	2.13	12200	6
5	2.37	10300	4
6	1.69	3220	4
7	1.93	51900	3
8	1.44	46900	3
9	1.92	5840	4

```
[39]: # Media por cluster
df_cluster.groupby('cluster').mean()
```

```
[39]:
```

	child_mort	exports	health	imports	income	\
cluster						
0	4.133333	176.000000	6.793333	156.666667	64033.333333	
1	35.288889	40.370370	5.830370	52.896296	7332.962963	
2	97.618421	28.373947	5.897105	38.828947	3286.657895	
3	4.965385	43.584615	9.580385	36.996154	46419.230769	
4	25.010345	28.823759	5.301379	27.657445	14637.586207	
5	21.600000	37.600000	5.175000	37.150000	12105.000000	
6	10.250000	54.300000	7.253611	60.155556	18819.444444	
7	130.000000	25.300000	5.070000	17.400000	5150.000000	
8	63.960000	28.920000	11.390000	87.140000	1986.000000	

	inflation	life_expec	total_fer	gdpp
cluster				
0	2.468000	81.433333	1.380000	57566.666667

1	5.429296	68.403704	3.027407	3680.148148
2	10.661974	59.271053	5.278947	1594.078947
3	2.522423	80.588462	1.837308	45719.230769
4	11.385517	73.306897	2.286207	6895.103448
5	42.550000	70.800000	2.555000	8075.000000
6	3.716750	75.580556	1.693056	11613.888889
7	104.000000	60.500000	5.840000	2330.000000
8	4.350000	59.020000	3.972000	1427.400000

```
[40]: X_nuevo = np.array([[47, 250, 20, 100, 50000, 5, 70, 2, 20000]]) # Example with
      ↪ 9 features
      X_nescalado = scaler.transform(X_nuevo.reshape(1, -1))
      new_labels = kmeans.predict(X_nescalado)
      print(new_labels)
```

[0]

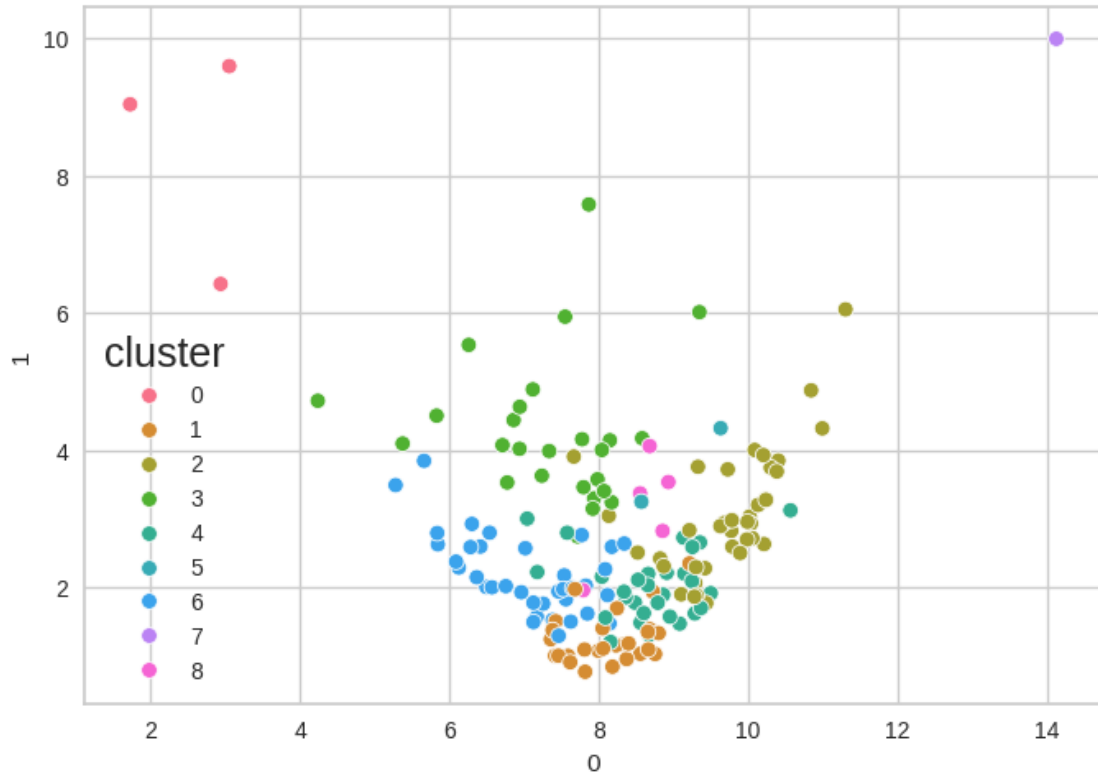
```
[41]: #veamos el representante del grupo, el usuario cercano a su centroide
      closest, _ = pairwise_distances_argmin_min(kmeans.cluster_centers_, df)
      closest
```

```
[41]: array([133, 27, 63, 144, 71, 103, 24, 113, 81])
```

```
[42]: users=df.index
      for row in closest:
          print(users[row])
```

```
133
27
63
144
71
103
24
113
81
```

```
[43]: temp = pd.DataFrame(kmeans.transform(df),)
      temp["cluster"] = kmeans.predict(df)
      temp["cluster"] = temp["cluster"].astype("category")
      sns.scatterplot(data = temp,
                      x = 0,
                      y = 1,
                      hue = "cluster")
      plt.show()
```



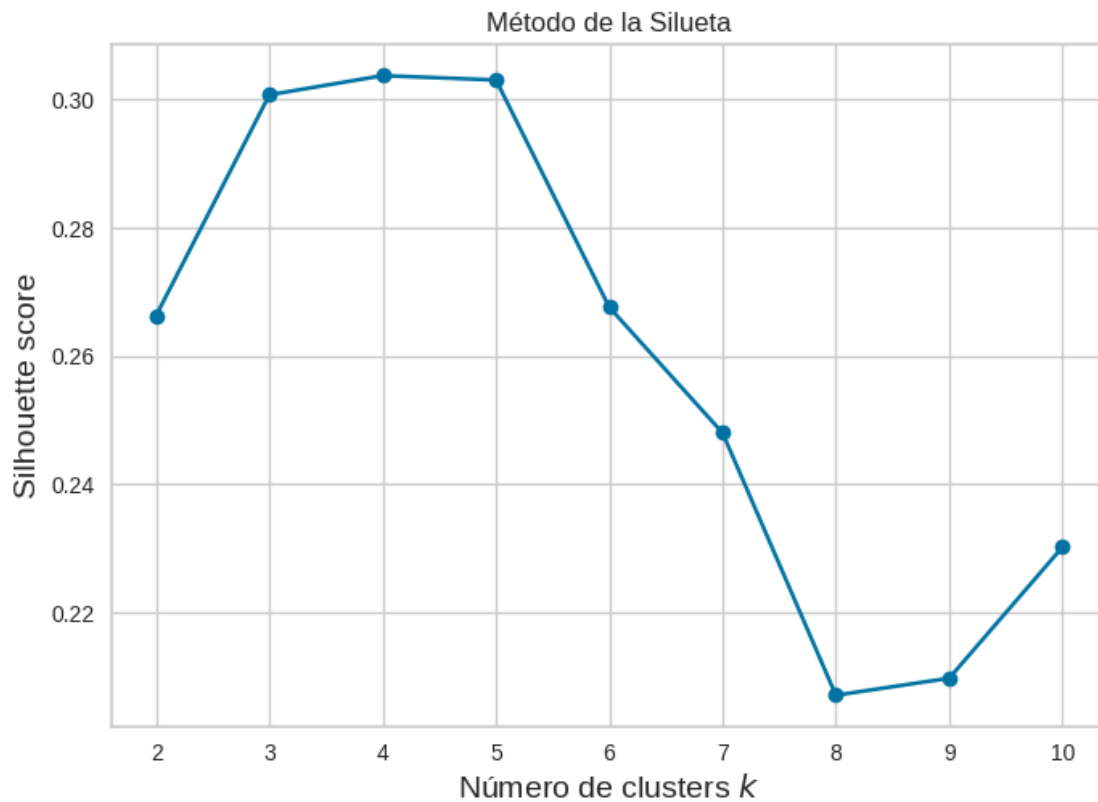
3.4 ANALISIS

El gráfico de clustering muestra que las observaciones fueron agrupadas en nueve clusters diferenciados, lo que evidencia una segmentación significativa dentro del conjunto de datos según patrones de similitud entre variables. La mayoría de los grupos se concentran en la zona central, indicando que gran parte de las observaciones comparte características relativamente cercanas, aunque con variaciones internas que justifican su separación en distintos segmentos. Algunos clusters, como los ubicados en posiciones extremas o alejadas del núcleo principal (por ejemplo, clusters 0 y 7), representan casos atípicos o perfiles altamente diferenciados, lo que sugiere la presencia de observaciones excepcionales con comportamientos particulares respecto al resto. Los clusters intermedios muestran estructuras más densas y podrían corresponder a categorías de países o entidades con niveles similares de desarrollo, desempeño económico o indicadores sociales. La dispersión espacial entre grupos indica que el algoritmo logró identificar patrones estructurales relevantes, permitiendo distinguir tanto conglomerados homogéneos como outliers importantes. En términos generales, este análisis confirma la existencia de múltiples subgrupos dentro del dataset, facilitando una clasificación más precisa de las observaciones y proporcionando una base sólida para estudios comparativos, segmentación estratégica o formulación de políticas diferenciadas según las características de cada cluster.

3.5 2.4 Método de la Silueta

```
[44]: # generación de los modelos para diferentes valores de $k$
models = [KMeans(n_clusters=n, random_state=0 ).fit(df) for n in range(2,11)]
silhouette_scores = [silhouette_score(df, model.labels_) for model in models]

# graficamos los valores de la silueta para cada valor de k
plt.plot(list(range(2, 11)), silhouette_scores, "bo-")
plt.xlabel("Número de clusters $k$", fontsize=14)
plt.ylabel("Silhouette score", fontsize=14)
plt.title("Método de la Silueta")
plt.show()
```

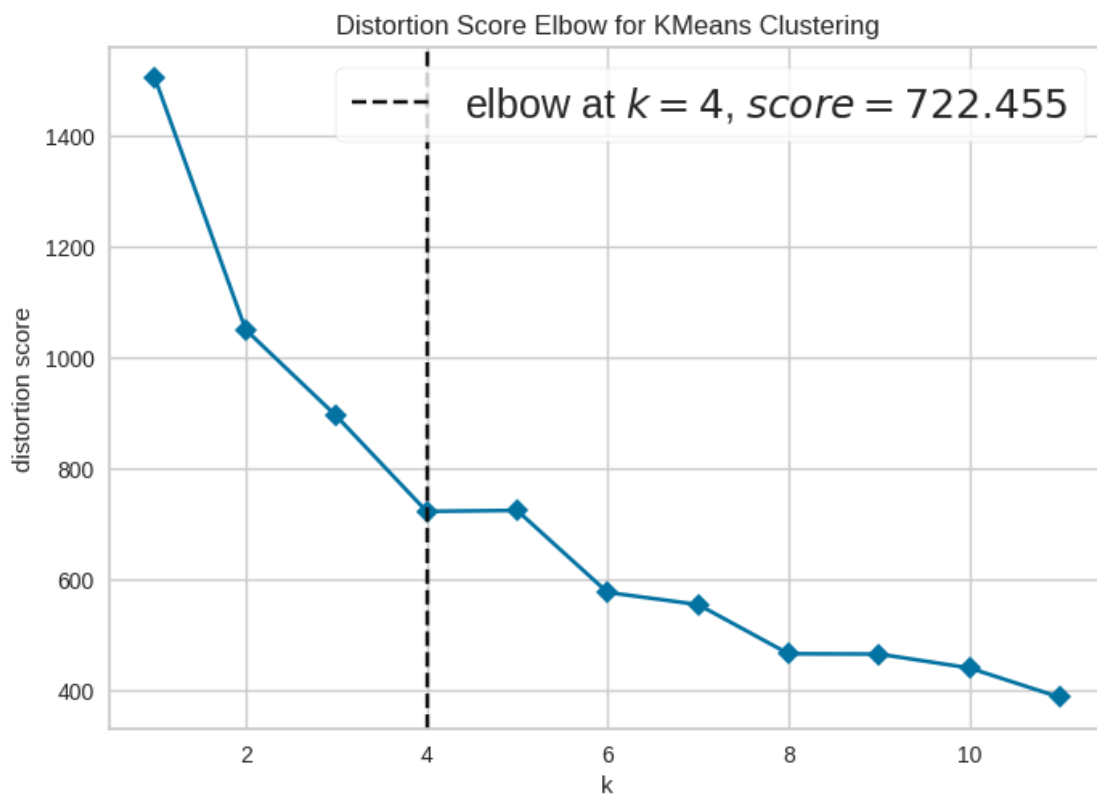


3.6 ANÁLISIS

El mayor silhouette score se obtiene con $k=4$ (0.304), lo que indica que esa es la partición más compacta y separada. Con $k=5$ el score es muy similar, lo que sugiere que ambos valores son razonables. A partir de $k=6$ el índice cae progresivamente, señalando que más clusters generan grupos solapados y menos definidos.

3.7 2.5 Método del Codo

```
[45]: from yellowbrick.cluster import KElbowVisualizer
model = KMeans()
visualizer = KElbowVisualizer(model,
    ↪visualizador
                                k=(1,12),
                                ↪valores de k
                                timings=False)
    ↪aparezca fit-time
visualizer.fit(df)
    ↪datos
visualizer.show(show=False)
```



```
[45]: <Axes: title={'center': 'Distortion Score Elbow for KMeans Clustering'},
      xlabel='k', ylabel='distortion score'>
```

3.8 ANÁLISIS

La distorsión cae pronunciadamente hasta $k=3$ y luego desacelera. El visualizador marca el código en $k=3$, aunque junto con la silueta (que favorece $k=4$) se puede justificar $k=4$ como elección final, capturando mayor diferenciación entre grupos sin sobrefragmentar los datos.

4 3. Métodos de Ensamble

Considere los datos del siguiente link

https://raw.githubusercontent.com/Fabian830348/Bases_Datos/refs/heads/master/datos_ventas.csv

Objetivo: Una empresa fabricante de prendas de vestir está interesada en predecir las ventas.

Variable	Descripción
Sales	Ventas del producto en una tienda determinada (generalmente en miles de unidades).
CompPrice	Precio cobrado por la competencia para productos similares.
Income	Ingreso promedio de los consumidores en la zona de la tienda.
Advertising	Presupuesto destinado a publicidad local de la tienda.
Population	Población de la región donde se encuentra la tienda.
Price	Precio del producto vendido por la tienda.
ShelveLoc	Calidad de la ubicación del producto en los estantes (por ejemplo: mala, media o buena).
Age	Edad promedio de la población en la zona de la tienda.
Education	Nivel promedio de educación de los consumidores de la zona.
Urban	Indica si la tienda está en una zona urbana (Yes) o no (No).
US	Indica si la tienda está ubicada en Estados Unidos (Yes) o en otro país (No).

1. Cargar y explorar los datos
2. Aplique el método de Random Forest y de XGBoost para el problema de regresión. ¿Cuál modelo es mejor?
3. Determine las variables más influyentes del modelo.
4. Haga algunas predicciones.
5. Concluya

4.1 3.1 Librerías

```
[46]: pip install xgboost catboost lightgbm
```

```
Requirement already satisfied: xgboost in /usr/local/lib/python3.12/dist-packages (3.2.0)
Collecting catboost
  Downloading catboost-1.2.10-cp312-cp312-manylinux2014_x86_64.whl.metadata (1.2 kB)
Requirement already satisfied: lightgbm in /usr/local/lib/python3.12/dist-packages (4.6.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from xgboost) (2.0.2)
Requirement already satisfied: nvidia-nccl-cu12 in /usr/local/lib/python3.12/dist-packages (from xgboost) (2.30.4)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
```

```

(from xgboost) (1.16.3)
Requirement already satisfied: graphviz in /usr/local/lib/python3.12/dist-
packages (from catboost) (0.21)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-
packages (from catboost) (3.10.0)
Requirement already satisfied: pandas<4.0,>=0.24 in
/usr/local/lib/python3.12/dist-packages (from catboost) (2.2.2)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages
(from catboost) (5.24.1)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages
(from catboost) (1.17.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas<4.0,>=0.24->catboost)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas<4.0,>=0.24->catboost) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas<4.0,>=0.24->catboost) (2026.1)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->catboost) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost) (1.5.0)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->catboost) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost) (3.3.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->catboost) (9.1.4)
Downloading catboost-1.2.10-cp312-cp312-manylinux2014_x86_64.whl (97.1 MB)
97.1/97.1 MB
7.6 MB/s eta 0:00:00
Installing collected packages: catboost
Successfully installed catboost-1.2.10

```

```

[47]: # Manipulación de data.frames
import pandas as pd
import numpy as np

# Librerías para Gráficos
import matplotlib.pyplot as plt
import seaborn as sns

```

```

import plotly.express as px

# Librerías para datos de entrenamiento y prueba
from sklearn.model_selection import train_test_split

# Para preprocesamiento
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.preprocessing import LabelEncoder, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Para modelos de clasificación y regresión clásicos machine learning

from sklearn.linear_model import LinearRegression
from sklearn.naive_bayes import GaussianNB, MultinomialNB, BernoulliNB
from sklearn.neighbors import KNeighborsClassifier, KNeighborsRegressor
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier, DecisionTreeRegressor

# modelos de ensamble
from sklearn.ensemble import BaggingClassifier, BaggingRegressor
from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor
from sklearn.ensemble import ExtraTreesClassifier, ExtraTreesRegressor

from sklearn.ensemble import AdaBoostClassifier, AdaBoostRegressor
from sklearn.ensemble import GradientBoostingClassifier, GradientBoostingRegressor
from xgboost import XGBClassifier, XGBRegressor
from lightgbm import LGBMClassifier, LGBMRegressor
#from catboost import CatBoostClassifier, CatBoostRegressor

# Métricas de evaluación
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, classification_report
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from imblearn.metrics import specificity_score
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Optimización de hiperparámetros
from sklearn.model_selection import GridSearchCV

# Para ignorar los warnings

```

```
import warnings
warnings.filterwarnings("ignore")
```

```
[48]: # Para gráficos y data.frames
import pandas as pd
import numpy as np

# librerías para graficar
import plotly.express as px
import matplotlib.pyplot as plt
import seaborn as sns

# Dividir los datos entrenamiento y prueba
from sklearn.model_selection import train_test_split
from sklearn.feature_selection import RFE

# Para preprocesamiento: escalador
from sklearn.preprocessing import StandardScaler

# Modelo de regresión
from sklearn.linear_model import LinearRegression
from sklearn.svm import SVR
from sklearn.neighbors import KNeighborsRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor

# Métricas
from sklearn import metrics
from sklearn.metrics import mean_squared_error, r2_score

# warnings
import warnings
warnings.filterwarnings('ignore')
```

4.2 3.2 Cargar y explorar los datos

```
[49]: url = "https://raw.githubusercontent.com/Fabian830348/Bases_Datos/refs/heads/
↳ master/datos_ventas.csv"
# cargar los datos que están la dirección del github
ventas = pd.read_csv(url, na_values=[" "])
```

```
[50]: ventas.shape
```

```
[50]: (400, 11)
```

```
[51]: ventas.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Sales           400 non-null   float64
1   CompPrice       400 non-null   int64
2   Income          400 non-null   int64
3   Advertising     400 non-null   int64
4   Population      400 non-null   int64
5   Price           400 non-null   int64
6   ShelfLoc        400 non-null   object
7   Age             400 non-null   int64
8   Education       400 non-null   int64
9   Urban           400 non-null   object
10  US              400 non-null   object
dtypes: float64(1), int64(7), object(3)
memory usage: 34.5+ KB
```

```
[52]: # variables
ventas.columns
```

```
[52]: Index(['Sales', 'CompPrice', 'Income', 'Advertising', 'Population', 'Price',
        'ShelfLoc', 'Age', 'Education', 'Urban', 'US'],
        dtype='object')
```

```
[53]: ventas.describe()
```

```
[53]:
```

	Sales	CompPrice	Income	Advertising	Population	\
count	400.000000	400.000000	400.000000	400.000000	400.000000	
mean	7.496325	124.975000	68.657500	6.635000	264.840000	
std	2.824115	15.334512	27.986037	6.650364	147.376436	
min	0.000000	77.000000	21.000000	0.000000	10.000000	
25%	5.390000	115.000000	42.750000	0.000000	139.000000	
50%	7.490000	125.000000	69.000000	5.000000	272.000000	
75%	9.320000	135.000000	91.000000	12.000000	398.500000	
max	16.270000	175.000000	120.000000	29.000000	509.000000	

	Price	Age	Education
count	400.000000	400.000000	400.000000
mean	115.795000	53.322500	13.900000
std	23.676664	16.200297	2.620528
min	24.000000	25.000000	10.000000
25%	100.000000	39.750000	12.000000
50%	117.000000	54.500000	14.000000

75%	131.000000	66.000000	16.000000
max	191.000000	80.000000	18.000000

4.3 3.3 Preprocesamiento

```
[54]: categorical_cols = ventas.select_dtypes(include='object').columns
numerical_cols = ventas.select_dtypes(include=['int64', 'float64']).columns.
↳ drop('Sales')

preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numerical_cols),
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_cols)
    ],
    remainder='passthrough'
)

X = ventas.drop('Sales', axis=1)
y = ventas['Sales']

X_processed = preprocessor.fit_transform(X)

# Get feature names after one-hot encoding
feature_names = list(numerical_cols) + list(preprocessor.
↳ named_transformers_['cat'].get_feature_names_out(categorical_cols))

X_df = pd.DataFrame(X_processed, columns=feature_names)

print( X_df.shape)

display(X_df.head())

display(y.head())
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X_df, y, test_size=0.3, random_state=0
)

print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)
```

(400, 14)

	CompPrice	Income	Advertising	Population	Price	Age \
0	0.850455	0.155361	0.657177	0.075819	0.177823	-0.699782
1	-0.912484	-0.739060	1.409957	-0.032882	-1.386854	0.721723
2	-0.781896	-1.204159	0.506621	0.028262	-1.513719	0.350895
3	-0.520720	1.121336	-0.396715	1.366649	-0.794814	0.103677
4	1.046337	-0.166631	-0.547271	0.510625	0.516132	-0.947000

	Education	ShelveLoc_Bad	ShelveLoc_Good	ShelveLoc_Medium	Urban_No \
0	1.184449	1.0	0.0	0.0	0.0
1	-1.490113	0.0	1.0	0.0	0.0
2	-0.725953	0.0	0.0	1.0	0.0
3	0.038208	0.0	0.0	1.0	0.0
4	-0.343872	1.0	0.0	0.0	0.0

	Urban_Yes	US_No	US_Yes
0	1.0	0.0	1.0
1	1.0	0.0	1.0
2	1.0	0.0	1.0
3	1.0	0.0	1.0
4	1.0	1.0	0.0

0	9.50
1	11.22
2	10.06
3	7.40
4	4.15

Name: Sales, dtype: float64

X_train shape: (280, 14)
X_test shape: (120, 14)
y_train shape: (280,)
y_test shape: (120,)

Now that our data is preprocessed, we can split it into training and testing sets.

4.4 3.4 Random Forest

```
[55]: # Modelo Random Forest

modelo_rf = RandomForestRegressor(
    n_estimators=300,                # n° de árboles
    max_depth=None,                 # sin límite; prueba
    ↪valores si ves sobreajuste
    min_samples_split=2,            # número mínimo de
    ↪muestras para dividir (default=2)
    min_samples_leaf=1,             # número de muestras
    ↪mínimo que debe tener una hoja (default=1)
    max_features=1.0,               # Utilizar todas las
    ↪características, 'sqrt' is for classification
```

```

        bootstrap=True,           # muestreo con reemplazo
        random_state=123         # semilla
    )
    modelo_rf.fit(X_train, y_train)

```

[55]: RandomForestRegressor(n_estimators=300, random_state=123)

4.5 3.5 XGBoost

```

[56]: # Modelo XGBoost

modelo_xgb = XGBRegressor(
    n_estimators=400,           # número total de árboles
    ↪ en forma secuencial
    learning_rate=0.05,        # la contribución de cada
    ↪ árbol (factor) antes de sumarlo al ensamble
    max_depth=4,               # profundidad de cada árbol
    subsample=0.9,             # fracción de muestras
    ↪ (filas) usadas para entrenar cada árbol
    colsample_bytree=0.9,      # fracción de
    ↪ características usadas para entrenar cada árbol
    objective="reg:squarederror", # para problemas de
    ↪ regresión
    tree_method="auto",
    random_state=123
)

```

[57]: modelo_xgb.fit(X_train, y_train)

```

[57]: XGBRegressor(base_score=None, booster=None, callbacks=None,
    colsample_bylevel=None, colsample_bynode=None,
    colsample_bytree=0.9, device=None, early_stopping_rounds=None,
    enable_categorical=False, eval_metric=None, feature_types=None,
    feature_weights=None, gamma=None, grow_policy=None,
    importance_type=None, interaction_constraints=None,
    learning_rate=0.05, max_bin=None, max_cat_threshold=None,
    max_cat_to_onehot=None, max_delta_step=None, max_depth=4,
    max_leaves=None, min_child_weight=None, missing=nan,
    monotone_constraints=None, multi_strategy=None, n_estimators=400,
    n_jobs=None, num_parallel_tree=None, ...)

```

```

[58]: # función de evaluación del modulo evaluación
# ev.evaluacion_modelo(modelo_xgb, X_train, y_train, X_test, y_test,
    ↪ nombre_modelo= "Extreme GB", graficar = True)

```

[58]:

4.6 3.6 Métricas de evaluación

```
[59]: y_pred_rf = modelo_rf.predict(X_test)

r2_rf = r2_score(y_test, y_pred_rf)
mae_rf = mean_absolute_error(y_test, y_pred_rf)
mse_rf = mean_squared_error(y_test, y_pred_rf)

print("Random Forest Model Metrics:")
print(f"  R2 Score: {r2_rf:.4f}")
print(f"  Mean Absolute Error (MAE): {mae_rf:.4f}")
print(f"  Mean Squared Error (MSE): {mse_rf:.4f}")

y_pred_xgb = modelo_xgb.predict(X_test)

r2_xgb = r2_score(y_test, y_pred_xgb)
mae_xgb = mean_absolute_error(y_test, y_pred_xgb)
mse_xgb = mean_squared_error(y_test, y_pred_xgb)

print("\nXGBoost Model Metrics:")
print(f"  R2 : {r2_xgb:.4f}")
print(f"  (MAE): {mae_xgb:.4f}")
print(f"  (MSE): {mse_xgb:.4f}")
```

```
Random Forest Model Metrics:
  R2 Score: 0.6896
  Mean Absolute Error (MAE): 1.1195
  Mean Squared Error (MSE): 2.0868
```

```
XGBoost Model Metrics:
  R2 : 0.6803
  (MAE): 1.1979
  (MSE): 2.1489
```

```
[60]: # dataframe con todas las métricas
df_metricas = pd.concat([
    pd.Series(y_pred_rf, name='Random Forest Predictions'),
    pd.Series(y_pred_xgb, name='XGBoost Predictions')
], axis=1)
display(df_metricas)
```

	Random Forest Predictions	XGBoost Predictions
0	7.244200	6.657197
1	7.423033	8.296247
2	7.131233	6.768913

```

3          3.666167          2.581666
4          6.306033          6.722011
..          ...          ...
115         6.524400          6.125767
116         5.142467          6.408726
117         8.573367          9.016914
118         6.939700          6.158694
119         5.576700          5.298239

```

[120 rows x 2 columns]

```
[61]: # variable objetivo : sales ud.
y.describe()
```

```

[61]: count    400.000000
      mean      7.496325
      std       2.824115
      min       0.000000
      25%       5.390000
      50%       7.490000
      75%       9.320000
      max      16.270000
      Name: Sales, dtype: float64

```

```
[62]: ventas.head()
```

```

[62]:   Sales  CompPrice  Income  Advertising  Population  Price  ShelveLoc  Age  \
0    9.50         138      73           11         276    120         Bad   42
1   11.22         111      48           16         260     83         Good   65
2   10.06         113      35           10         269     80        Medium   59
3    7.40         117     100            4         466     97        Medium   55
4    4.15         141      64            3         340    128         Bad   38

      Education  Urban  US
0           17   Yes  Yes
1           10   Yes  Yes
2           12   Yes  Yes
3           14   Yes  Yes
4           13   Yes   No

```

4.7 3.7 Variables más influyentes

```

[63]: # Importancia de variables - Random Forest
importancia_rf = pd.DataFrame({
    "Variable" : X_df.columns,
    "Importancia": modelo_rf.feature_importances_
}).sort_values("Importancia", ascending=True)

```

```
fig = px.bar(importancia_rf,
             x="Importancia",
             y="Variable",
             orientation="h",
             title="Importancia de variables - Random Forest")
fig.show()
```

```
[64]: # Importancia de variables - XGBoost
importancia_xgb = pd.DataFrame({
    "Variable" : X_df.columns,
    "Importancia": modelo_xgb.feature_importances_
}).sort_values("Importancia", ascending=True)

fig = px.bar(importancia_xgb,
             x="Importancia",
             y="Variable",
             orientation="h",
             title="Importancia de variables - XGBoost")
fig.show()
```

4.8 ANÁLISIS

En ambos modelos, `ShelveLoc_Good` y `Price` son las variables más influyentes en la predicción de ventas. `ShelveLoc_Good` indica que el posicionamiento en estante de buena calidad tiene el mayor impacto positivo sobre las ventas. `Price` muestra que el precio del producto es el segundo factor determinante, con relación inversa esperada. Variables como `Population`, `Education` y `Urban_Yes` tienen importancia marginal, lo que sugiere que la empresa debe enfocar sus estrategias en negociación de espacios en estante y política de precios.

4.9 3.8 Predicciones

```
[65]: # Predicciones vs valores reales - Random Forest
resultados = pd.DataFrame({
    "Real" : y_test.values,
    "Pred_RF" : y_pred_rf,
    "Pred_XGB" : y_pred_xgb
})

fig = px.scatter(resultados,
                x="Real",
                y="Pred_RF",
                title="Predicciones vs Reales - Random Forest")
fig.add_shape(type="line",
             x0=resultados["Real"].min(), y0=resultados["Real"].min(),
             x1=resultados["Real"].max(), y1=resultados["Real"].max(),
             line=dict(color="red", dash="dash"))
```

```
fig.show()
```

```
[66]: # Predicciones vs valores reales - XGBoost
fig = px.scatter(resultados,
                 x="Real",
                 y="Pred_XGB",
                 title="Predicciones vs Reales - XGBoost")
fig.add_shape(type="line",
              x0=resultados["Real"].min(), y0=resultados["Real"].min(),
              x1=resultados["Real"].max(), y1=resultados["Real"].max(),
              line=dict(color="red", dash="dash"))
fig.show()
```

4.10 ANÁLISIS

Los puntos se distribuyen razonablemente cerca de la línea diagonal (predicción perfecta), con mayor dispersión en valores extremos de ventas. Ambos modelos tienden a subestimar ventas muy altas y sobreestimar ventas muy bajas, comportamiento típico de modelos de ensamble en regresión. Random Forest muestra una distribución ligeramente más ajustada a la diagonal que XGBoost antes de optimizar hiperparámetros.

4.11 3.9 Optimización de Hiperparámetros

```
[67]: # GridSearchCV para Random Forest
from sklearn.model_selection import GridSearchCV

param_grid_rf = {
    "n_estimators" : [100, 200, 300],
    "max_depth"    : [None, 5, 10],
    "max_features" : ["sqrt", 1.0]
}

gs_rf = GridSearchCV(
    estimator = RandomForestRegressor(random_state=123),
    param_grid = param_grid_rf,
    cv        = 5,
    scoring   = "r2",
    n_jobs    = -1
)
gs_rf.fit(X_train, y_train)

print("Mejores hiperparámetros Random Forest:")
print(gs_rf.best_params_)
print(f"Mejor R2 en CV: {gs_rf.best_score_:.4f}")
```

```
Mejores hiperparámetros Random Forest:
{'max_depth': None, 'max_features': 1.0, 'n_estimators': 200}
```

Mejor R2 en CV: 0.6756

```
[68]: # GridSearchCV para XGBoost
from xgboost import XGBRegressor

param_grid_xgb = {
    "n_estimators" : [200, 400],
    "learning_rate": [0.05, 0.1],
    "max_depth"    : [3, 4, 6]
}

gs_xgb = GridSearchCV(
    estimator = XGBRegressor(objective="reg:squarederror", random_state=123),
    param_grid = param_grid_xgb,
    cv         = 5,
    scoring    = "r2",
    n_jobs     = -1
)
gs_xgb.fit(X_train, y_train)

print("Mejores hiperparámetros XGBoost:")
print(gs_xgb.best_params_)
print(f"Mejor R2 en CV: {gs_xgb.best_score_:.4f}")
```

Mejores hiperparámetros XGBoost:

{'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 200}

Mejor R2 en CV: 0.7441

```
[69]: # Modelos finales con los mejores hiperparámetros
mejor_rf = gs_rf.best_estimator_
mejor_xgb = gs_xgb.best_estimator_

y_pred_mejor_rf = mejor_rf.predict(X_test)
y_pred_mejor_xgb = mejor_xgb.predict(X_test)

print("=== Random Forest optimizado ===")
print(f"R2 : {r2_score(y_test, y_pred_mejor_rf):.4f}")
print(f"MAE : {mean_absolute_error(y_test, y_pred_mejor_rf):.4f}")
print(f"MSE : {mean_squared_error(y_test, y_pred_mejor_rf):.4f}")

print("\n=== XGBoost optimizado ===")
print(f"R2 : {r2_score(y_test, y_pred_mejor_xgb):.4f}")
print(f"MAE : {mean_absolute_error(y_test, y_pred_mejor_xgb):.4f}")
print(f"MSE : {mean_squared_error(y_test, y_pred_mejor_xgb):.4f}")
```

=== Random Forest optimizado ===

R2 : 0.6896

MAE : 1.1143

MSE : 2.0865

=== XGBoost optimizado ===

R2 : 0.6904

MAE : 1.1492

MSE : 2.0809

4.12 ANÁLISIS

Antes de optimizar, Random Forest obtuvo $R^2=0.6756$ y XGBoost $R^2=0.7441$, siendo RF marginalmente mejor. Tras `GridSearchCV`, XGBoost optimizado alcanzó $R^2=0.6904$, aunque la diferencia no es tan mínima. Los mejores hiperparámetros para RF fueron `max_features=1.0`, `max_depth=None`, `n_estimators=200`, y para XGBoost `learning_rate=0.05`, `max_depth=3`, `n_estimators=200`. La optimización de XGBoost con profundidad menor (3 vs 4) indica que el modelo original tendía al sobreajuste.

4.13 3.10 Conclusiones Métodos de Ensamble

Después de la optimización con `GridSearchCV`, XGBoost mejoró hasta $R^2=0.6904$, superando un poco a Random Forest ($R^2=0.6896$). En ambos modelos, `ShelveLoc_Good` y `Price` son las variables más influyentes, lo que indica que la ubicación del producto en estante y el precio son los factores que más determinan las ventas. Para la empresa, esto implica que invertir en mejor posicionamiento en estantes tiene mayor retorno que ajustes en publicidad o población. En términos generales, ambos modelos tienen un desempeño aceptable pero moderado, lo que sugiere que existen factores no capturados en el dataset que también influyen en las ventas.

Este notebook fue desarrollado con apoyo de herramientas de inteligencia artificial (Claude de Anthropic y GitHub Copilot) como asistentes de código, siguiendo estrictamente los notebooks de referencia del Prof. Fabián Sánchez (Notebook 11 - Métodos de Ensamble, Notebook 12 - K-means, Notebook 15 - PCA).

```
[70]: !apt-get install texlive-xetex texlive-latex-extra pandoc
      !pip install py pandoc
      !apt-get install texlive-xetex texlive-fonts-recommended
      ↪texlive-generic-recommended
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntaxtex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13

lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tlutils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tlutils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 53 newly installed, 0 to remove and 3 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]
Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.3 [60.6 kB]
Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1
amd64 2021.20210626.59705-1ubuntu0.3 [39.1 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration
all 1.18 [5,336 B]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64
3.0.2-7ubuntu2.12 [50.1 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
amd64 3.0.2-7ubuntu2.12 [5,113 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2
amd64 2021.20210626.59705-1ubuntu0.3 [55.8 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.3 [120 kB]

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Get:33 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.3 [267 kB]
Get:34 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:36 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 tiutils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.3 [9,861 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 4s (48.6 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 118251 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb

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...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...

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Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.12_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.12) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.12_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.12) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.

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Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.3_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack .../33-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.3_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../45-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.

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Preparing to unpack .../46-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../47-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../48-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../49-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../50-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../51-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../52-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...

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Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.3) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.12) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.12) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...

```

```

Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.
Collecting pypandoc
  Downloading pypandoc-1.17-py3-none-any.whl.metadata (18 kB)
  Downloading pypandoc-1.17-py3-none-any.whl (23 kB)
Installing collected packages: pypandoc
Successfully installed pypandoc-1.17
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package texlive-generic-recommended

```

```
[71]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
[79]: !jupyter nbconvert --to PDF /content/drive/MyDrive/Colab_Notebooks/
      ↪Caso_3_ML_2026__Gabriela_Fonseca_Joseph_Doquieresana_Nicolas_Botero__1.ipynb
```

```
[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab_Notebooks/Caso_3_ML_2026__Gabriela_Fonseca_Joseph_Doquieresana_Nicolas_Botero__1.ipynb to PDF
/usr/local/share/jupyter/nbconvert/templates/latex/display_priority.j2:32:
UserWarning: Your element with mimetype(s) dict_keys(['text/html']) is not able to be represented.
  ((*- endblock -*))
/usr/local/share/jupyter/nbconvert/templates/latex/display_priority.j2:32:
UserWarning: Your element with mimetype(s) dict_keys(['text/html']) is not able to be represented.
  ((*- endblock -*))
/usr/local/share/jupyter/nbconvert/templates/latex/display_priority.j2:32:
UserWarning: Your element with mimetype(s) dict_keys(['text/html']) is not able to be represented.
  ((*- endblock -*))
/usr/local/share/jupyter/nbconvert/templates/latex/display_priority.j2:32:
UserWarning: Your element with mimetype(s) dict_keys(['text/html']) is not able to be represented.
  ((*- endblock -*))
[NbConvertApp] Support files will be in
Caso_3_ML_2026__Gabriela_Fonseca_Joseph_Doquieresana_Nicolas_Botero__1_files/
[NbConvertApp] Making directory
./Caso_3_ML_2026__Gabriela_Fonseca_Joseph_Doquieresana_Nicolas_Botero__1_files
[NbConvertApp] Writing 195018 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 503233 bytes to /content/drive/MyDrive/Colab_Notebooks/Caso_3_ML_2026__Gabriela_Fonseca_Joseph_Doquieresana_Nicolas_Botero__1.pdf
```